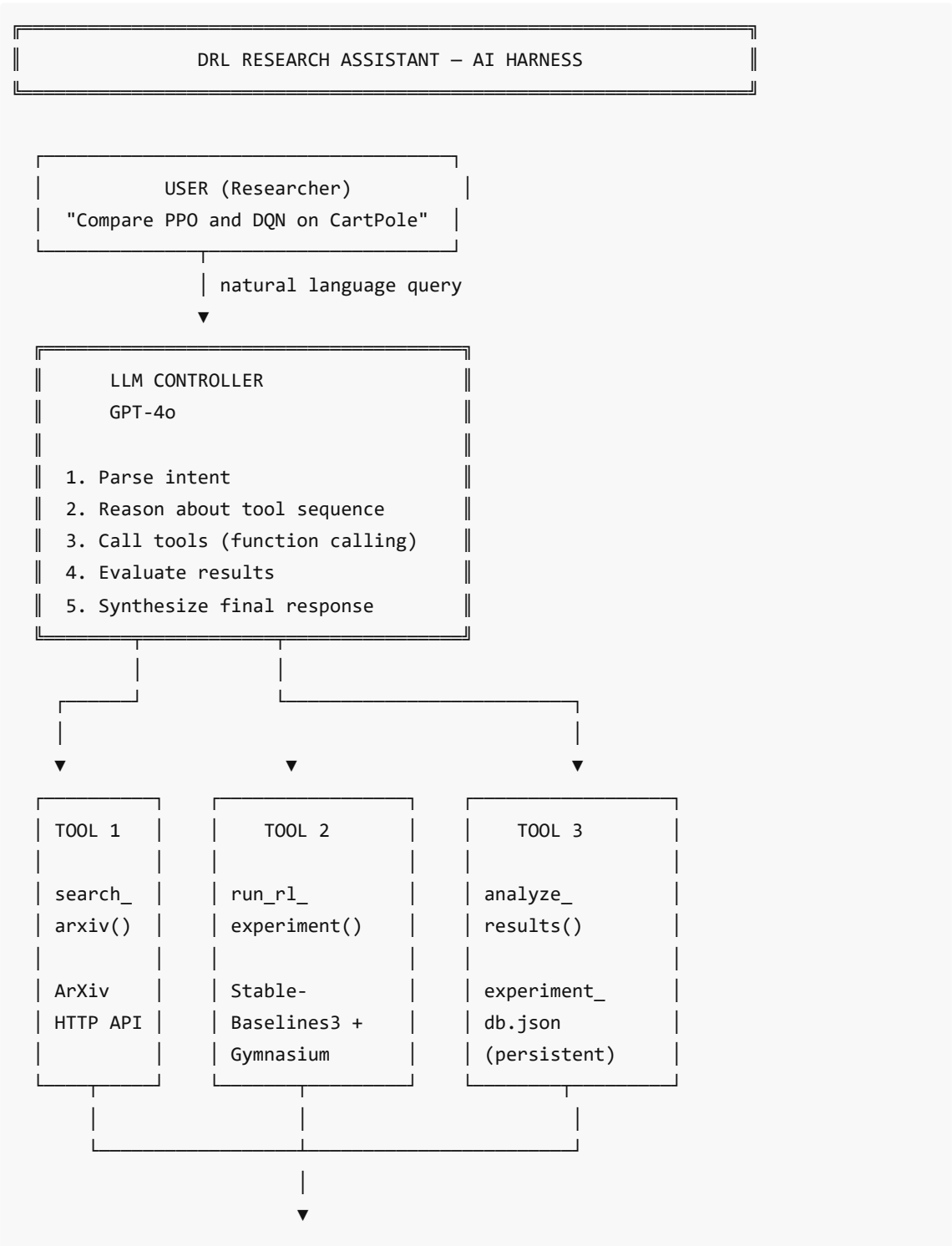
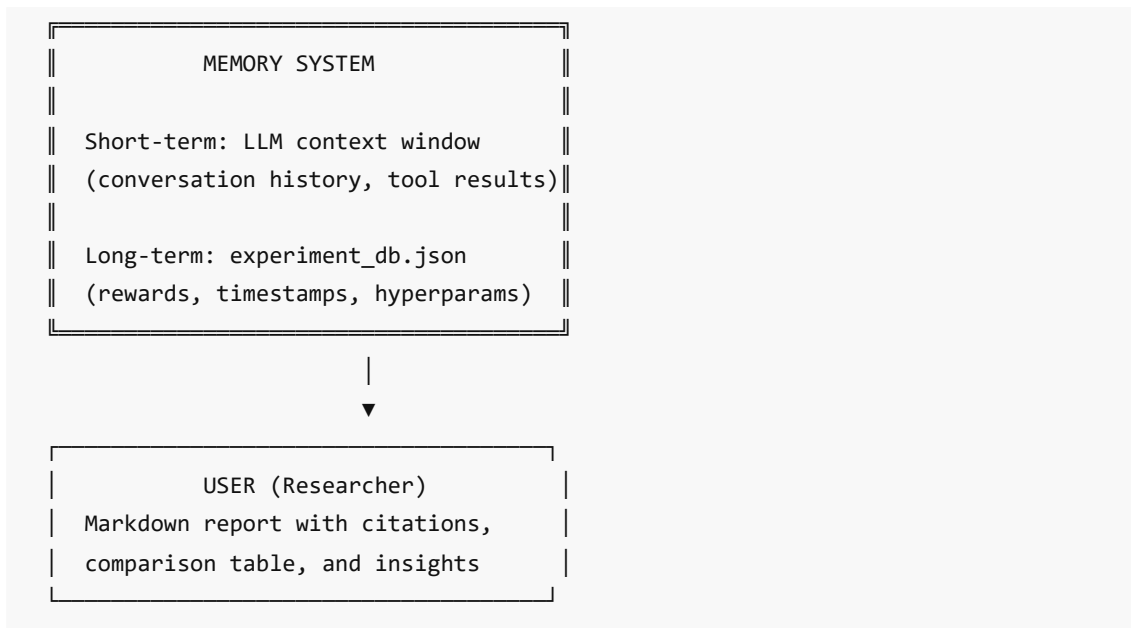


# DRL Research Assistant — System Infographic

Render this file in any Markdown viewer (GitHub, Obsidian, VS Code).  
Mermaid diagrams render natively on GitHub and in most modern editors.

## 1. System Architecture Overview





## 2. Function Calling / Tool Chain Flow (Sequence Diagram)

```
sequenceDiagram
    participant U as User
    participant LLM as LLM Controller<br/>(GPT-4o)
    participant T1 as Tool 1<br/>search_arxiv
    participant T2 as Tool 2<br/>run_rl_experiment
    participant T3 as Tool 3<br/>analyze_results
    participant DB as experiment_db.json

    U->>LLM: "Compare PPO and DQN on CartPole (50k steps)"

    Note over LLM: Reasoning: need literature first
    LLM->>T1: search_arxiv(query="DQN PPO CartPole", max_results=3)
    T1-->>LLM: [{title, authors, abstract, url}, ...]

    Note over LLM: Reasoning: run PPO experiment
    LLM->>T2: run_rl_experiment(algorithm="PPO", environment="CartPole-v1",
n_steps=50000)
    T2->>DB: save result {exp_id: "a1b2", mean_reward: 487.3}
    T2-->>LLM: {experiment_id: "a1b2", mean_reward: 487.3, status: "completed"}

    Note over LLM: Reasoning: run DQN experiment
    LLM->>T2: run_rl_experiment(algorithm="DQN", environment="CartPole-v1",
n_steps=50000)
    T2->>DB: save result {exp_id: "c3d4", mean_reward: 144.1}
    T2-->>LLM: {experiment_id: "c3d4", mean_reward: 144.1, status: "completed"}
```

Note over LLM: Reasoning: compare both runs

LLM->>T3: analyze\_results(experiment\_ids=["a1b2", "c3d4"])

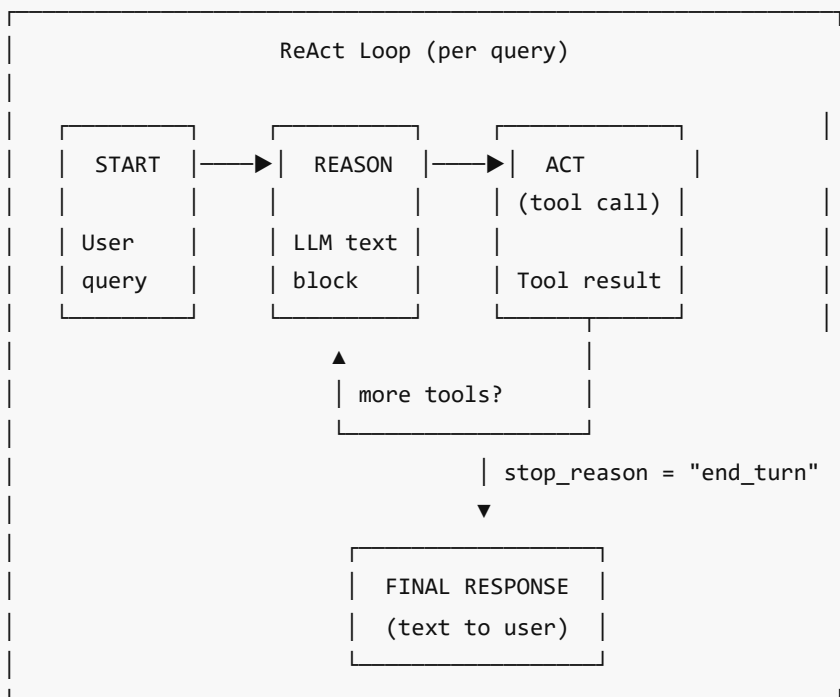
T3->>DB: load experiments "a1b2", "c3d4"

T3-->>LLM: {best: PP0@487.3, table: [...], insights: [...]}

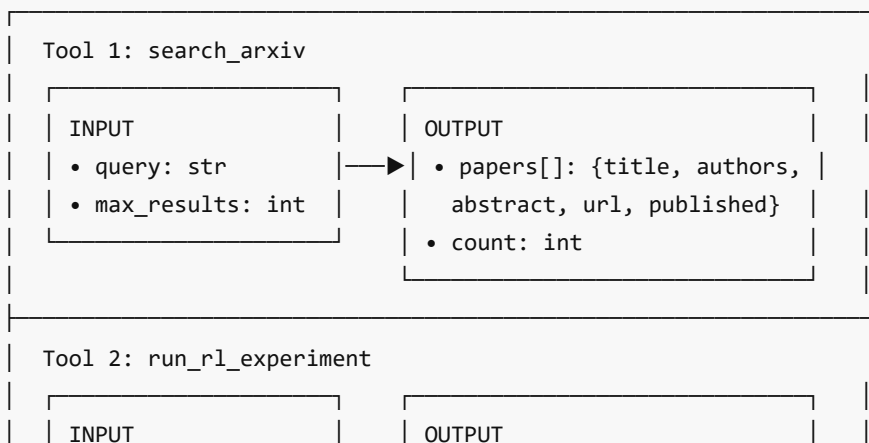
Note over LLM: Synthesize final response

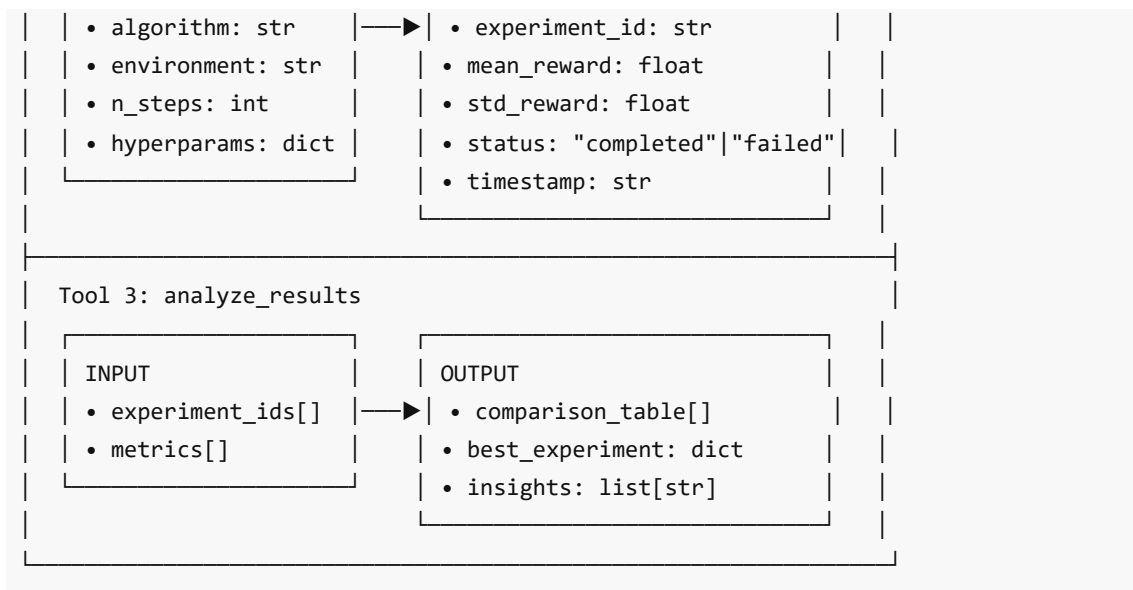
LLM-->>U: Markdown report with citations, comparison table, and recommendation

### 3. ReAct Orchestration Loop



### 4. Tool Input / Output Summary





## 5. Memory Architecture



